

Evidence matrix

Requested capability	Evidence in checkout	Documentation treatment
YOLOv8 detection	Settings page names YOLOv8-X/N, but no model-loading or inference call exists; <code>models/yolov11.pt</code> is present.	Proposed production architecture; current prototype uses OpenCV Canny/contours.
Defect inspection	<code>pages/5_?_AI_Inspection.py</code> performs grayscale conversion, Gaussian blur, Canny edges, contour filtering, pixel-to-mm conversion and rule-based severity.	Implemented and mathematically documented.
3D twin	<code>pages/7_??_3D_Building.py</code> contains a parameterized Three.js viewer; <code>app.py</code> contains a Plotly frame viewer.	Implemented visual prototype.
Structural screening	<code>modules/digital_twin/engine.py</code> calculates tributary gravity, axial stress, simple beam moment/stress, wind pressure, hydrostatic pressure and utilization.	Implemented preliminary screening only.
4D seismic	UI label <code>Seismic Oscillation</code> exists, but no seismic equation or time-history solver is implemented.	Proposed extension with equations, explicitly not a measured result.
ESG carbon forecast	No carbon/emission logic found in source.	Proposed lifecycle-accounting extension, not a project result.
IoT/telemetry	Demo sensor data in page 3; deterministic synthetic telemetry in <code>engine.py</code> ; typed telemetry contracts in <code>platform_core/contracts.py</code> .	Implemented demo/contracts; production ingestion remains future work.
Cost estimation	<code>pages/9_?_Cost_Estimation.py</code> implements material, labour, scaffolding, GST and quantity calculations.	Implemented formulae reproduced.
Solver integration	<code>workers/dispatcher.py</code> checks COLMAP/OpenSees/OpenFOAM availability and returns READY/BLOCKED.	Implemented readiness boundary; no solver output fabricated.

Project analysis summary

ConstructVision AI is best understood as an **inspection-workflow and decision-support prototype**. Its immediate strength is not a trained artificial-intelligence model or a certified structural analysis result. Its strength is the attempt to keep a field observation, its approximate measurement, its asset context, a transparent preliminary check, a recommended workflow and a report in one place.

The checkout has two user-interface layers. The root `app.py` provides the newer operational command centre: portfolio, twin studio, capture and reconstruction, health, telemetry and integration views. The `pages/` directory remains a richer prototype library for tutorial, virtual practical, GPS, inspection, damage analysis, 3D building, IoT, cost, materials, history, reports and configuration. `platform_core/` and `workers/` provide a deliberately small API-oriented foundation for a future service architecture.

Verified source inventory

Area	Primary source files	What is demonstrably present	Important boundary
Application shell	<code>main.py</code> , <code>app.py</code> , <code>modules/portfolio.py</code>	Streamlit entry point, operational navigation, Plotly frame visualisation and command-centre screens.	Presentation is local/prototype; no production authentication boundary is evidenced here.
Image triage	<code>pages/5_?_AI_Inspection.py</code> , <code>modules/ai_detection.py</code> , <code>modules/inspection/</code>	Image upload, OpenCV-style inspection workflow and rule-based severity support.	A contour/heuristic result is not a calibrated defect diagnosis.
Structural context	<code>pages/7_??_3D_Building.py</code> , <code>modules/digital_twin/engine.py</code> , <code>modules/building/</code>	Parametric geometry, frame nodes/members and simple mechanical indicators.	No code-compliant design, FEA result or calibrated as-built model is established.

Area	Primary source files	What is demonstrably present	Important boundary
Telemetry	pages/8_?_IoT.py, modules/iot/monitor.py, modules/sensor_state.py, platform_core/contracts.py	Demonstration channels, synthetic telemetry and validation models for telemetry records.	No live gateway, device identity, time-series database or alert-delivery evidence is recorded.
Repair workflow	pages/6_?_Damage_Analysis.py, pages/9_?_Cost_Estimation.py, modules/cost/	Repair/risk presentation, BOQ-style calculation and cost components.	Costs are planning estimates; site measurement, supplier quotation and engineer approval remain required.
Reporting and history	pages/11_?_History.py, pages/12_?_Reports.py, modules/reports.py, modules/history/	User-facing inspection history and report-generation workflow.	Immutable evidence storage, signatures and retention enforcement are future controls.
External analysis	workers/dispatcher.py	Explicit COLMAP, OpenSees and OpenFOAM readiness checks.	BLOCKED means no analysis has run; no result should be inferred.

Claim-control register

Claim category	Wording allowed in the current documentation	Wording to avoid until validation exists	Evidence needed to promote the claim
AI defect detection	"Rule-based/OpenCV-assisted image triage"	"YOLOv8 detects defects with X% accuracy"	Versioned dataset, model checkpoint, held-out evaluation and error analysis.
Crack dimensions	"Approximate dimensions derived from a user-supplied scale"	"Metric crack measurement"	Calibrated capture protocol, reference targets and measurement-error study.
3D twin	"Parametric visual twin"	"As-built BIM/digital twin"	Survey control, reconstruction report, model revision and spatial accuracy results.
Structural result	"Preliminary transparent screening indicator"	"Safe", "approved", "code compliant" or "design verified"	Governing design basis, load combinations, validated model and licensed engineer sign-off.
Sensor monitoring	"Synthetic/demo telemetry and typed record contract"	"Live monitoring" or "real-time alerts"	Device inventory, gateway logs, timestamp checks, data-quality report and alert audit trail.
Cost estimate	"Illustrative BOQ-style estimate"	"Final project cost"	Measured quantities, approved scope, vendor rates, tax treatment and approval record.

Evidence package required for a field pilot

Before a field pilot, retain a per-inspection evidence package containing the following items:

- Asset identifier, site/location reference, component identifier and inspection purpose.
- Original image/video files, capture timestamp, camera/device details and consent/access record.
- Calibration target or scale method, estimated measurement uncertainty and any image pre-processing record.
- Detector or rule-set revision, thresholds, raw outputs, reviewed outputs and reviewer identity.
- Sensor identifier, unit, time zone/UTC timestamp, quality flag, missing-data handling and calibration status.
- Structural-screening inputs, units, assumptions, result revision and the explicit preliminary-result warning.
- Repair recommendation, cost assumptions, exclusions, approvals and final report version.

Documentation maintenance rules

- Refer to the actual current page names: pages/5_?_AI_Inspection.py, pages/7_??_3D_Building.py and pages/9_?_Cost_Estimation.py.
- Put measured results in a dated test record; keep targets and illustrative values labelled as targets.

- Update this matrix whenever a model, dataset, sensor connection, solver worker or report-signing workflow changes.
- Preserve the distinction between implemented, demo/synthetic, readiness-only and proposed capabilities in every presentation and report.